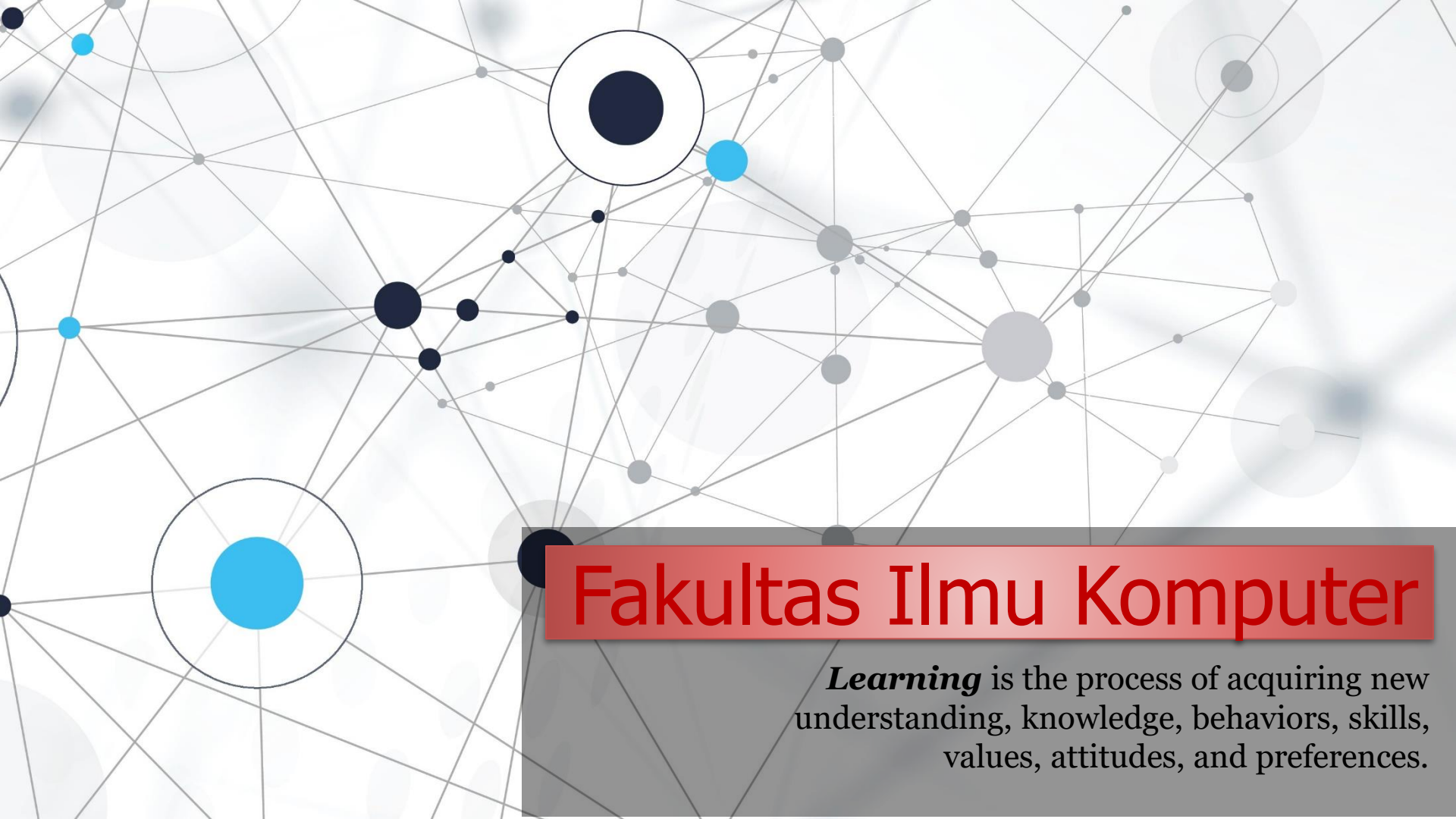


An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Create Read Update Delete (CRUD)



Welcome *to the Back-end Web Development course*

phpMyAdmin x Active Class x

https://unkpresent.site/unklab/public/dosen/active/

Universitas Klabat Search Sign out

Semmy Wellem Taju Active Classes Wednesday, 17 April 2024 10:07 AM

GENERAL MENU

- Home
- Informations
- Classroom Topics
- Logout

SHORTCUT LECTURER

- Active Classes
- Archived Classes
- Attendance List
- Review WF Status
- Magic Cut

Course	Meetings/Room	Schedule	Status	Actions
[IS2243] Object Oriented Programming - A Semester Genap 2023/2024 3 SKS (7 absences max limit) Fakultas Ilmu Komputer Taju, Semmy Wellem	36 Students 22 Public Scheduled GA - 203 Computer Lab	Selasa-Kamis 08.40 - 10.10	active	Options
[IF3262] Back-End Web Development - C Semester Genap 2023/2024 3 SKS (7 absences max limit) Fakultas Ilmu Komputer Taju, Semmy Wellem	8 Students 24 Public Scheduled GA - 203 Computer Lab	Senin-Rabu 10.10 - 11.40		
[IF3262] Back-End Web Development - D Semester Genap 2023/2024 3 SKS (7 absences max limit) Fakultas Ilmu Komputer Taju, Semmy Wellem	27 Students 23 Public Scheduled GA - 203 Computer Lab	Senin-Rabu 14.40 - 16.10		
[IS2243] Object Oriented Programming - G Semester Genap 2023/2024 3 SKS (7 absences max limit) Fakultas Ilmu Komputer Taju, Semmy Wellem	33 Students 22 Public Scheduled GK2 Computer Lab 2	Selasa-Kamis 16.10 - 17.30	active	Options

Info

- Add Students
- Attendance Schedules
- Archive Class
- Review WF Students
- Class Summary

https://unkpresent.site/unklab/public/students/attendance/16552

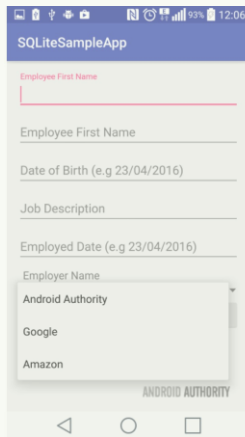
Bagaimana jika project kalian sudah terlalu banyak fungsi yang harus di handle??? Bagaimana cara pengaturannya?

A decorative border of green leaves and branches at the top of the slide.

1. What is CRUD ?

A decorative border of green leaves and branches at the bottom of the slide.

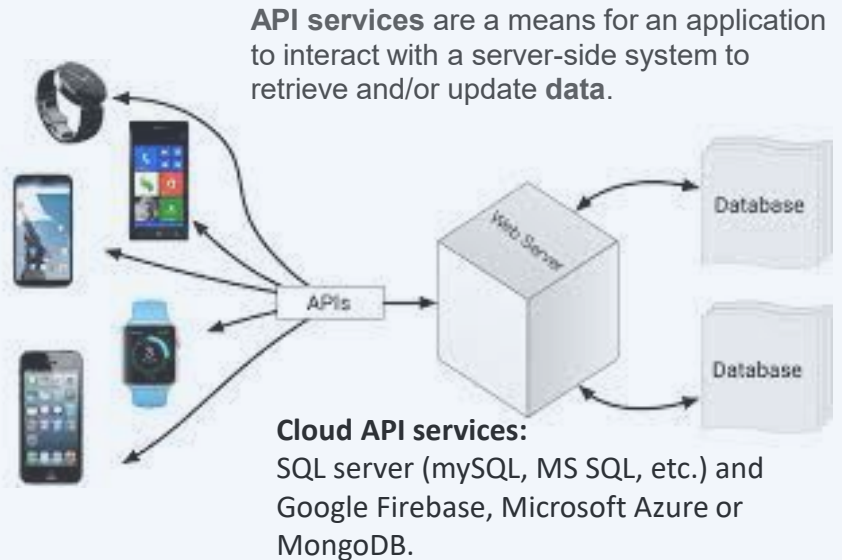
Types of Databases



**File on SD
(Secure Digital)
cards**



Local databases are usually faster than remote database servers because they reside on the same *system* as the database application.



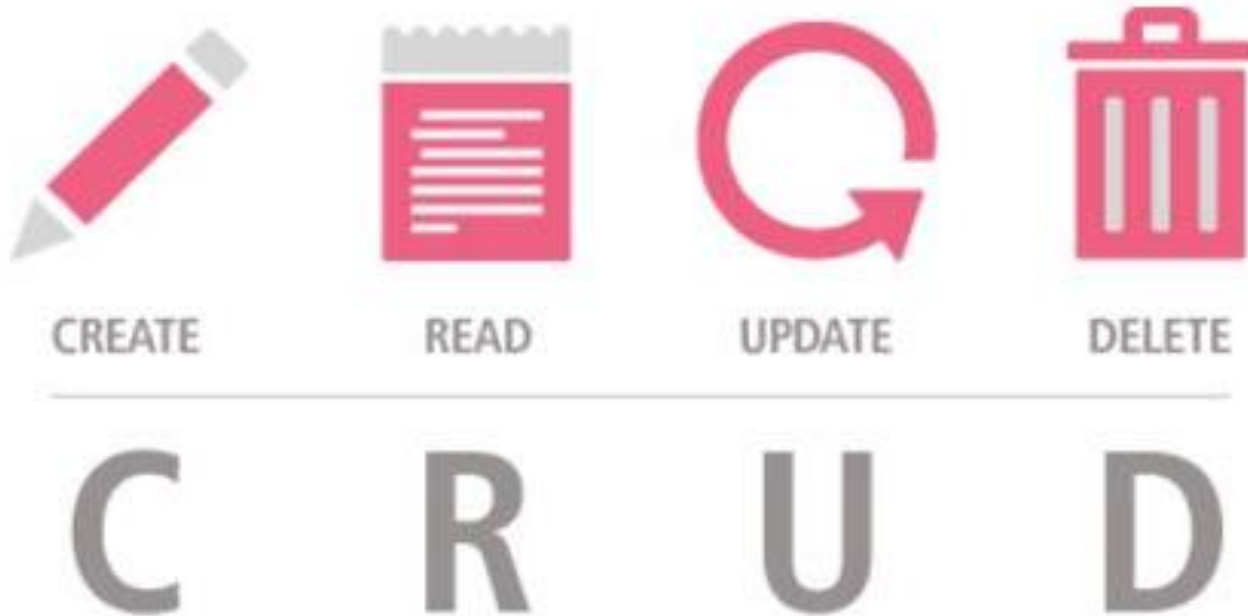
Remote **database servers** are optimized to support different types of operations, so we may want to consider performance when choosing a remote database *server*.

Today, MySQL Database Tutorial with AIMVC Framework

Insert, Delete, Update and View Data in/from Database

Apa itu CRUD Website?

CRUD adalah singkatan yang berasal dari Create, Read, Update dan Delete, dimana keempat istilah tersebut merupakan fungsi utama yang nantinya diimplementasikan ke dalam basis data.



CRUD Android

**Membuat
CRUD
dengan
database
SQLite di
Android
Studio**

← Splash

Name

Surname

Marks

Id

ADD DATA VIEW ALL

UPDATE DELETE

← Splash

Data

Id :4
Name :gibson
Surname :tyres
Marks :99

Id :7
Name :cggggyyh
Surname :ggggyg
Marks :vghvty

Id :9
Name :jhon
Surname :die
Marks :read

UPDATE DELETE

WHAT IS



CREATE

C



READ

R



UPDATE

U



DELETE

D

CREATE

Create



DELETE

READ



UPDATE

item 4 | Update

A decorative border of green leaves and branches at the top of the slide.

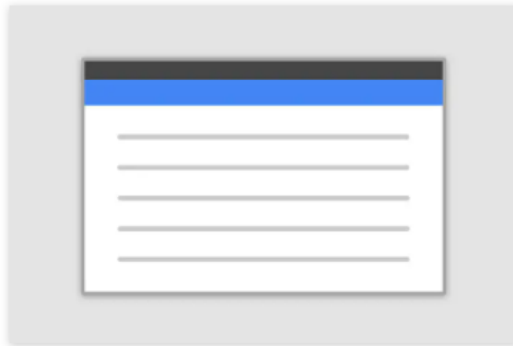
2. Modal Box, Input Form, GET & POST Methods

A decorative border of green leaves and branches at the bottom of the slide.

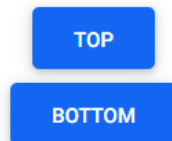
Bootstrap 5 Modal Box Component

Responsive modal built with the latest Bootstrap 5. **Modal is a responsive popup used to display extra content.** That includes prompts, configurations, cookie consents, *etc.*

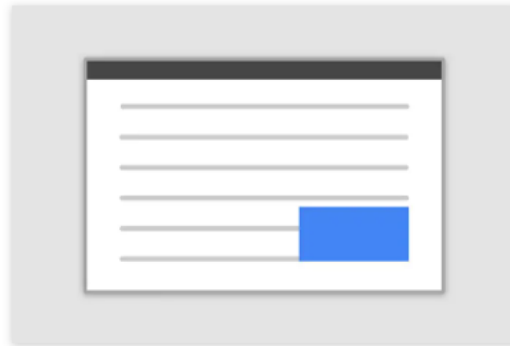
Frame modal



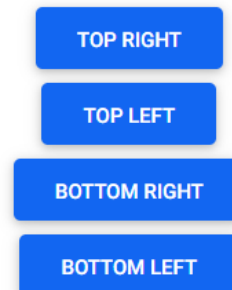
Position



Side modal



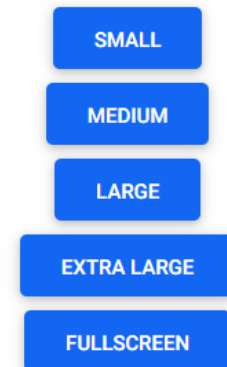
Position



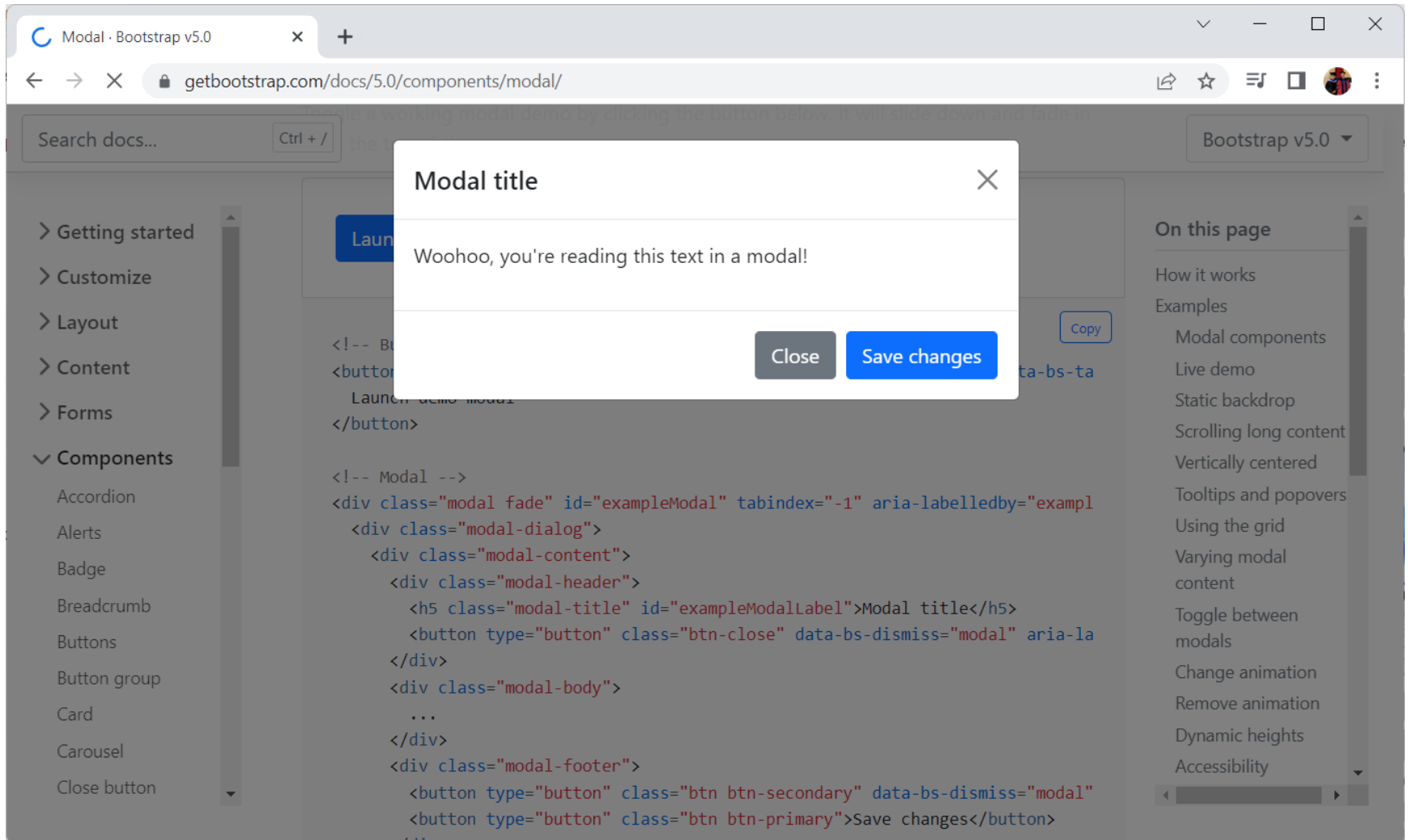
Central modal



Size



Bootstrap's Modal Component



Link: <https://getbootstrap.com/docs/5.0/components/modal/>

Bootstrap's Centered Modal

Bootstrap 5 Modal

w3schools.com/bootstrap5/bootstrap_modal.php

HTML CSS JAVASCRIPT SQL PYTHON PHP **BOOTSTRAP** HOW TO W3.CSS JAVA

BS5 Typography

BS5 Colors

BS5 Tables

BS5 Images

BS5 Jumbotron

BS5 Alerts

BS5 Buttons

BS5 Button Groups

BS5 Badges

BS5 Progress Bars

BS5 Spinners

BS5 Pagination

BS5 List Groups

BS5 Cards

BS5 Dropdowns

BS5 Collapse

BS5 Navs

BS5 Navbar

BS5 Carousel

BS5 Modal

BS5 Tooltip

BS5 Popover

BS5 Toast

Centered Modal

Center the modal vertically and horizontally within the page, with the `.modal-dialog-centered` class:

Example

```
<div class="modal-dialog modal-dialog-centered">
```

Try it Yourself »

Scrolling Modal

When you have alot of content inside the modal, a scrollbar is added to the page. See the examples below to understand it:

Example

ADVERTISEMENT

Skill Academy by ProKenja

Pengguna Skill Academy Mencapai > 10 Juta, Lho!

Jumlahnya **LEBIH BANYAK** dari populasi negara Int



DAFTAR KELAS SEKARANG!

Waiting for c.amazon-adsystem.com...

Bootstrap's Centered Modal

Tryit Editor v3.7

+

← → ↺

w3schools.com/bootstrap5/tryit.asp?filename=trybs_modal_centered

🔗 ☆ ☰ 🖼️ 👤 ⋮

🏠 ☰ 📁 🔄 🌙

Run >

Result Size: 602 x 511

Get your own website

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1">
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/css/bootstrap
.min.css" rel="stylesheet">
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/js/bootstrap.b
undle.min.js"></script>
</head>
<body>

<div class="container mt-3">
  <h3>Vertically Centered Modal Example</h3>
  <p>Click on the button to open the modal.</p>

  <button type="button" class="btn btn-primary" data-bs-
toggle="modal" data-bs-target="#myModal">
    Open modal
  </button>
</div>

<!-- The Modal -->
<div class="modal" id="myModal">
  <div class="modal-dialog modal-dialog-centered">
```

Vertically Centered Modal Example

Click on the button to open the modal.

Open modal

Modal Heading

Modal body..

Close

Bootstrap 5 Forms

This form will be used to input new data into database.

The screenshot displays the w3schools.com website's Bootstrap 5 Forms section. The browser's address bar shows the URL `w3schools.com/bootstrap5/bootstrap_forms.php`. The navigation menu at the top includes links for HTML, CSS, JAVASCRIPT, SQL, PYTHON, PHP, **BOOTSTRAP**, HOW TO, W3.CSS, and JAVA. The left sidebar lists various Bootstrap 5 components, with 'Bootstrap 5 Forms' and its sub-item 'BS5 Forms' highlighted. The main content area features the title 'Bootstrap 5 Forms' and two green buttons: '< Previous' and 'Next >'. Below this, the 'Stacked Form' section explains that all textual `<input>` and `<textarea>` elements with the class `.form-control` receive proper styling. An example form is shown with the following elements:

- Email:
- Password:
- ☐ Remember me
-

A 'NEW' button is visible in the bottom right corner of the page.

PHP - GET & POST Methods

What is \$_GET and \$_POST in PHP?

\$_GET variable is an array of variables passed to the current script via the URL parameters.

\$_POST variable is an array of variables passed to the current script via the HTTP POST method.

```
<!DOCTYPE HTML>
<html>
<body>

<form action="welcome.php" method="post">
Name: <input type="text" name="name"><br>
E-mail: <input type="text" name="email"><br>
<input type="submit">
</form>

</body>
</html>
```

```
<html>
<body>

Welcome <?php echo $_POST["name"]; ?><br>
Your email address is: <?php echo $_POST["email"]; ?>

</body>
</html>
```

```
<html>
<body>

<form action="welcome_get.php" method="get">
Name: <input type="text" name="name"><br>
E-mail: <input type="text" name="email"><br>
<input type="submit">
</form>

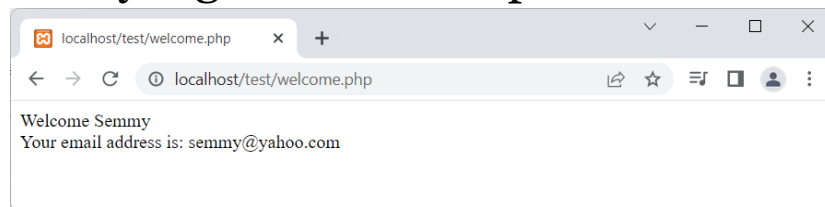
</body>
</html>
```

```
<html>
<body>

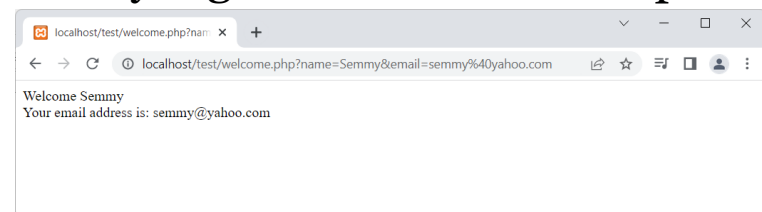
Welcome <?php echo $_GET["name"]; ?><br>
Your email address is: <?php echo $_GET["email"]; ?>

</body>
</html>
```

Data yang dikirim encrypted.



Data yang dikirim tdk di encrypt.



A decorative border of green leaves and branches at the top of the slide.

3. Last Meeting

(Component View Demo)

A decorative border of green leaves and branches at the bottom of the slide.

AIMVC Framework Structure

The screenshot shows a Notepad++ window titled 'C:\xampp\htdocs\lecture27\aimvc\app\init.php - Notepad++'. The interface includes a menu bar (File, Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, Plugins, Window, ?) and a toolbar. The 'Folder as Workspace' pane on the left shows the project structure:

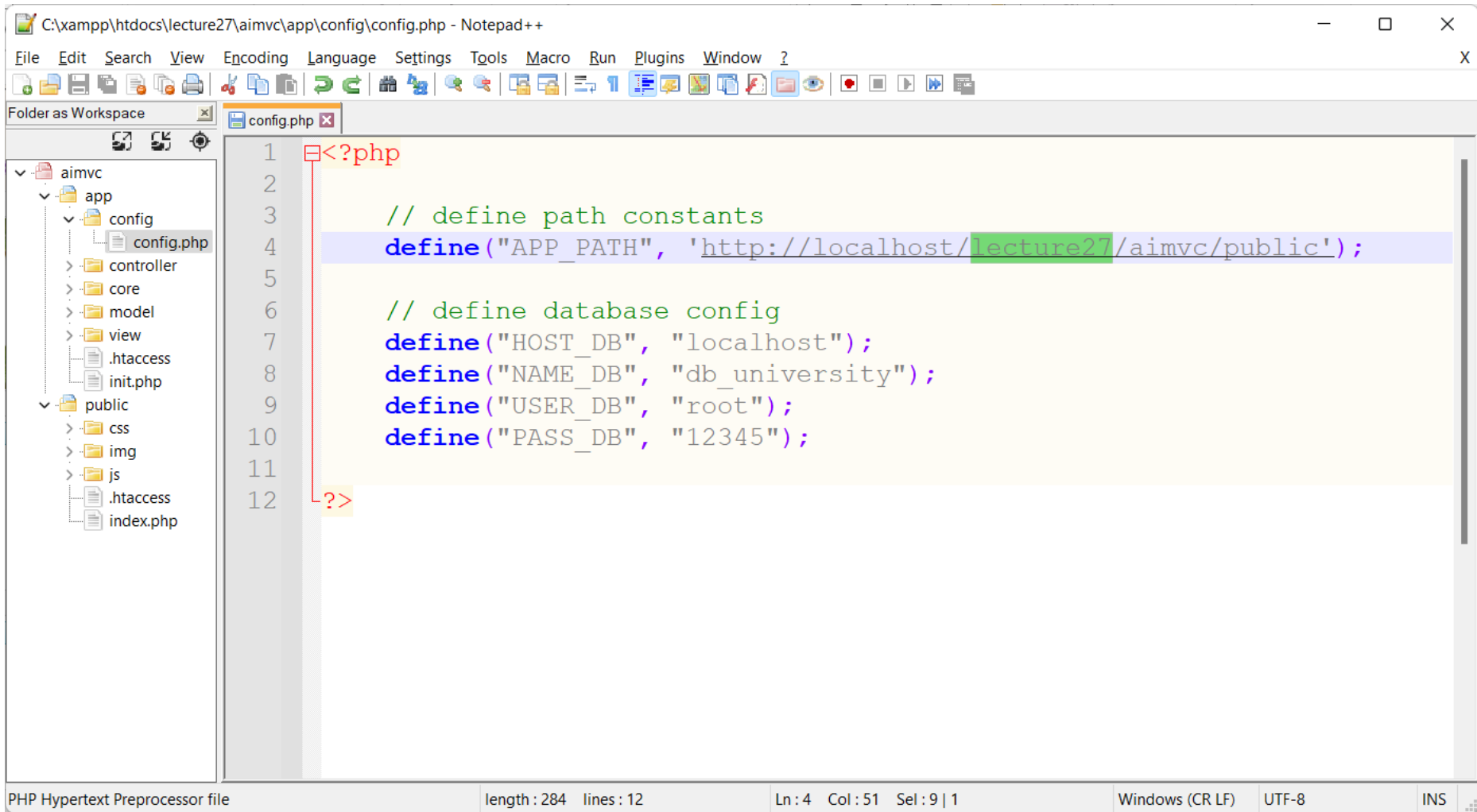
- aimvc
 - app
 - config
 - config.php
 - controller
 - Home.php
 - Project.php
 - core
 - App.php
 - Controller.php
 - Database.php
 - model
 - Lecturer_model.php
 - view
 - home
 - project
 - template
 - .htaccess
 - init.php
 - public
 - css
 - img
 - js
 - .htaccess
 - index.php

The main editor area displays the content of 'init.php' with line numbers 1 through 13. The code is as follows:

```
1 <?php
2     // load core classes
3     require_once('core/App.php');
4     require_once('core/Controller.php');
5     require_once('core/Database.php');
6
7     // load configuration file
8     require_once('config/config.php');
9
10    // start session
11    //session_start();
12    ?>
13
```

The status bar at the bottom indicates: 'PHP Hypertext Preprocessor file', 'length : 262 lines : 13', 'Ln : 1 Col : 1 Pos : 1', 'Windows (CR LF)', 'UTF-8', and 'INS'.

Don't Forget Root Path !!!



```
1 <?php
2
3 // define path constants
4 define("APP_PATH", 'http://localhost/lecture27/aimvc/public');
5
6 // define database config
7 define("HOST_DB", "localhost");
8 define("NAME_DB", "db_university");
9 define("USER_DB", "root");
10 define("PASS_DB", "12345");
11
12 ?>
```

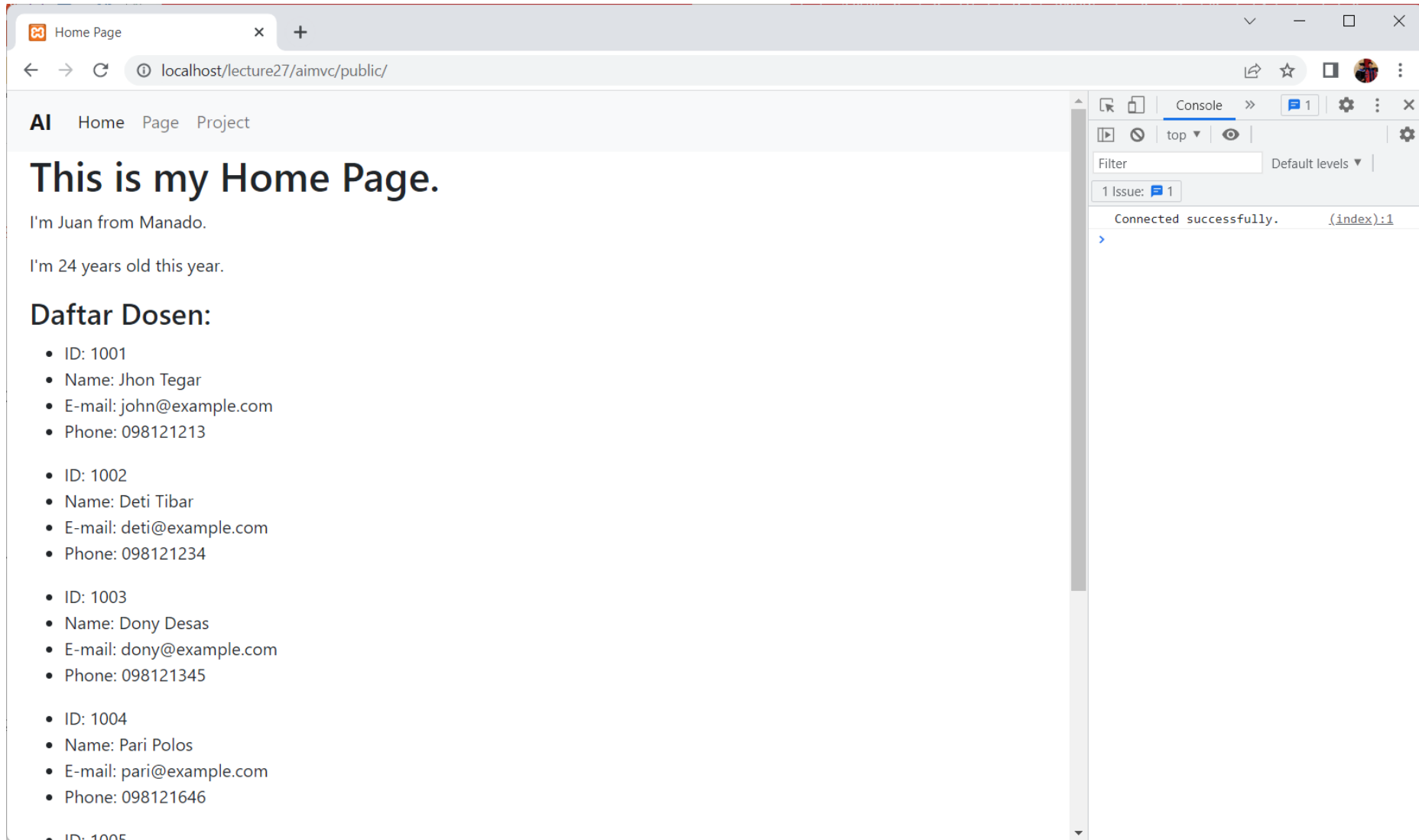
Folder as Workspace

- aimvc
 - app
 - config
 - config.php
 - controller
 - core
 - model
 - view
 - .htaccess
 - init.php
 - public
 - css
 - img
 - js
 - .htaccess
 - index.php

length : 284 lines : 12 Ln : 4 Col : 51 Sel : 9 | 1 Windows (CR LF) UTF-8 INS

I've already created new directory for today lecture.
Please remember also to modify this line of code if you move to the server side.

Read Data (Display Data *into* Browser)



The screenshot shows a web browser window with a single tab titled "Home Page". The address bar displays "localhost/lecture27/aimvc/public/". The page content includes a navigation bar with "AI", "Home", "Page", and "Project" links. The main heading is "This is my Home Page.", followed by two lines of text: "I'm Juan from Manado." and "I'm 24 years old this year." Below this is a section titled "Daftar Dosen:" containing a list of five teachers, each with their ID, Name, E-mail, and Phone number. On the right side of the browser, the developer console is open, showing a message "Connected successfully." with a source of "(index):1".

Home Page

localhost/lecture27/aimvc/public/

AI Home Page Project

This is my Home Page.

I'm Juan from Manado.

I'm 24 years old this year.

Daftar Dosen:

- ID: 1001
- Name: Jhon Tegar
- E-mail: john@example.com
- Phone: 098121213

- ID: 1002
- Name: Deti Tibar
- E-mail: deti@example.com
- Phone: 098121234

- ID: 1003
- Name: Dony Desas
- E-mail: dony@example.com
- Phone: 098121345

- ID: 1004
- Name: Pari Polos
- E-mail: pari@example.com
- Phone: 098121646

- ID: 1005

Console

1 Issue: 1

Connected successfully. (index):1



You're Started Here !!!



Exercise *for* Students



4. Display Data in Bootstrap 5 Tables

Bootstrap 5 Tables

Bootstrap 5 Tables

w3schools.com/bootstrap5/bootstrap_tables.php

Tutorials▼References▼Exercises▼VideosProNEWWebsiteNEWPaid CoursesLog in

HTMLCSSJAVASCRIPTSQLPYTHONPHPBOOTSTRAPHOW TOW3.CSSJAVASearch

Bootstrap 5 Tutorial

BS5 HOME

BS5 Get Started

BS5 Containers

BS5 Grid Basic

BS5 Typography

BS5 Colors

BS5 Tables

BS5 Images

BS5 Jumbotron

BS5 Alerts

BS5 Buttons

BS5 Button Groups

BS5 Badges

BS5 Progress Bars

BS5 Spinners

BS5 Pagination

BS5 List Groups

BS5 Cards

Bootstrap 5 Tables

< PreviousNext >

Basic Table

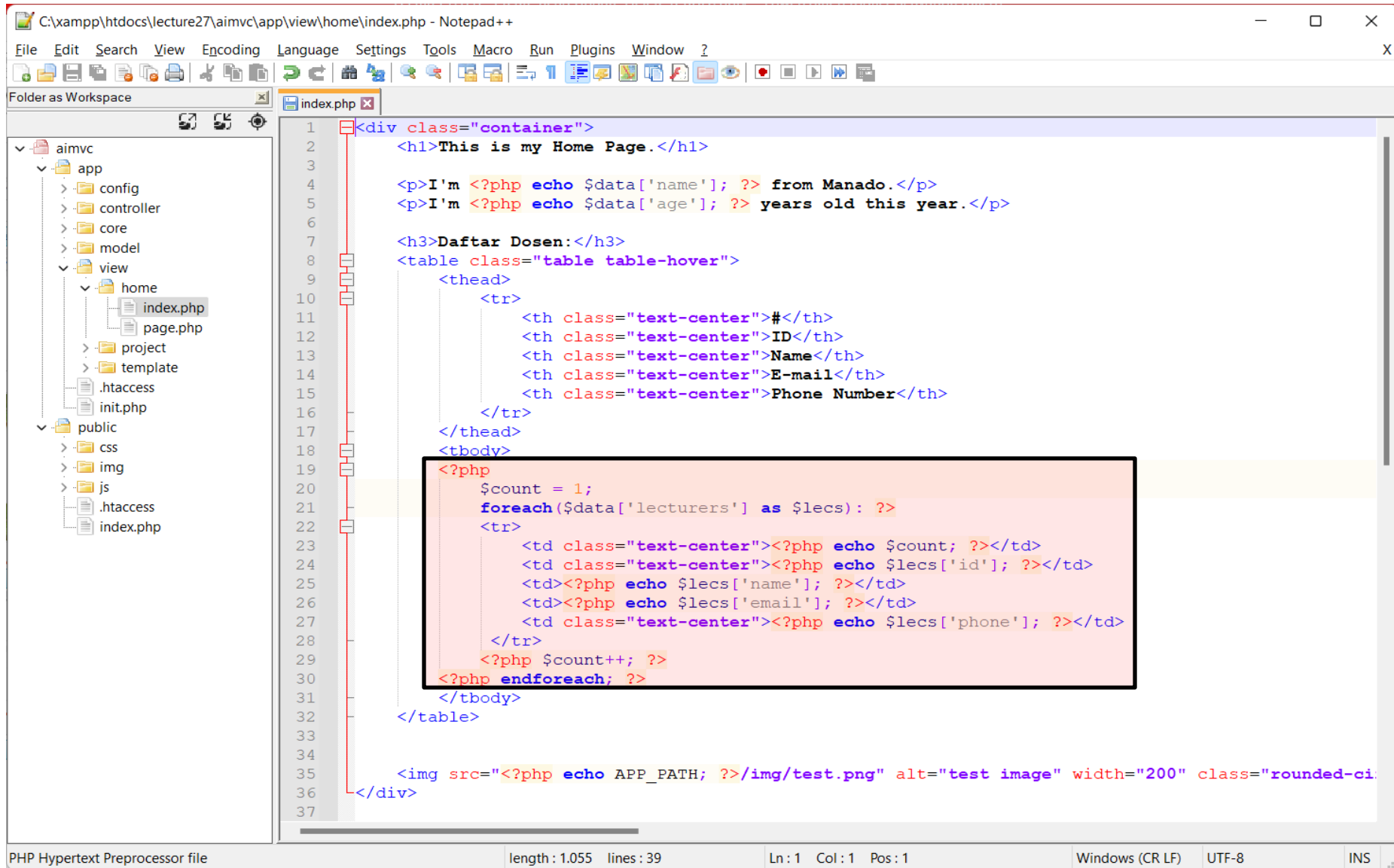
A basic Bootstrap 5 table has a light padding and horizontal dividers.

The `.table` class adds basic styling to a table:

Example

Lastname	Email

Modify Code -> view/home/index.php



The screenshot shows a Notepad++ window editing the file `C:\xampp\htdocs\lecture27\aimvc\app\view\home\index.php`. The file explorer on the left shows the project structure, with the `view/home/index.php` file selected. The code in the editor is as follows:

```
1 <div class="container">
2   <h1>This is my Home Page.</h1>
3
4   <p>I'm <?php echo $data['name']; ?> from Manado.</p>
5   <p>I'm <?php echo $data['age']; ?> years old this year.</p>
6
7   <h3>Daftar Dosen:</h3>
8   <table class="table table-hover">
9     <thead>
10      <tr>
11        <th class="text-center">#</th>
12        <th class="text-center">ID</th>
13        <th class="text-center">Name</th>
14        <th class="text-center">E-mail</th>
15        <th class="text-center">Phone Number</th>
16      </tr>
17    </thead>
18    <tbody>
19      <?php
20        $count = 1;
21        foreach($data['lecturers'] as $lecs): ?>
22          <tr>
23            <td class="text-center"><?php echo $count; ?></td>
24            <td class="text-center"><?php echo $lecs['id']; ?></td>
25            <td><?php echo $lecs['name']; ?></td>
26            <td><?php echo $lecs['email']; ?></td>
27            <td class="text-center"><?php echo $lecs['phone']; ?></td>
28          </tr>
29          <?php $count++; ?>
30        <?php endforeach; ?>
31      </tbody>
32    </table>
33
34    
36
37
```

The status bar at the bottom indicates: PHP Hypertext Preprocessor file, length: 1.055 lines: 39, Ln: 1 Col: 1 Pos: 1, Windows (CR LF), UTF-8, INS.

View Data in Table

Home Page

localhost/lecture27/aimvc/public/

AI Home Page Project

This is my Home Page.

I'm Juan from Manado.

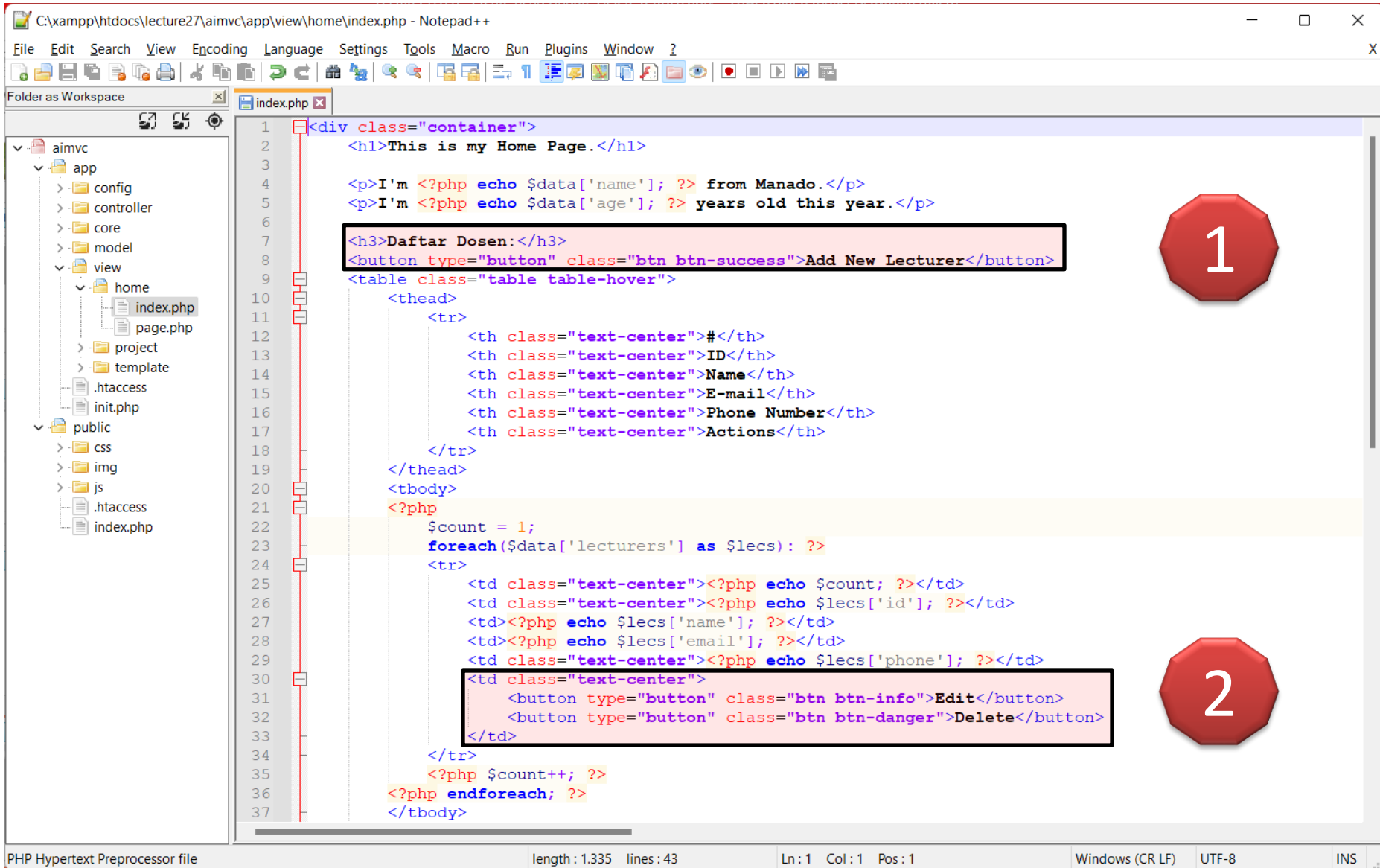
I'm 24 years old this year.

Daftar Dosen:

#	ID	Name	E-mail	Phone Number
1	1001	Jhon Tegar	john@example.com	098121213
2	1002	Deti Tibar	deti@example.com	098121234
3	1003	Dony Desas	dony@example.com	098121345
4	1004	Pari Polos	pari@example.com	098121646
5	1005	Soei Siali	soei@example.com	098121890



Add Buttons (Insert, Update & Delete)



The screenshot shows a Notepad++ editor window with the file path `C:\xampp\htdocs\lecture27\aimvc\app\view\home\index.php`. The editor displays PHP code for a web page. The code includes a header section, a table of lecturers, and a footer section. Two red octagonal callouts with numbers 1 and 2 highlight specific button elements in the code.

Callout 1 highlights the button element:

```
<button type="button" class="btn btn-success">Add New Lecturer</button>
```

Callout 2 highlights the button elements:

```
<button type="button" class="btn btn-info">Edit</button>
<button type="button" class="btn btn-danger">Delete</button>
```

The code also includes a table structure for displaying lecturer information, with columns for ID, Name, E-mail, Phone Number, and Actions.

PHP Hypertext Preprocessor file | length: 1.335 | lines: 43 | Ln: 1 | Col: 1 | Pos: 1 | Windows (CR LF) | UTF-8 | INS

Add Buttons (Insert, Update & Delete)

Home Page

localhost/lecture27/aimvc/public/

AI Home Page Project

This is my Home Page.

I'm Juan from Manado.

I'm 24 years old this year.

Daftar Dosen:

Add New Lecturer

#	ID	Name	E-mail	Phone Number	Actions	
1	1001	Jhon Tegar	john@example.com	098121213	Edit	Delete
2	1002	Deti Tibar	deti@example.com	098121234	Edit	Delete
3	1003	Dony Desas	dony@example.com	098121345	Edit	Delete
4	1004	Pari Polos	pari@example.com	098121646	Edit	Delete
5	1005	Soei Siali	soei@example.com	098121890	Edit	Delete



A decorative border of green leaves and branches at the top of the slide.

5. Insert, Update & Delete Data

(You need to change following files)

A decorative border of green leaves and branches at the bottom of the slide.

File #1: /app/core/Database.php

```
1 <?php
2 class Database{
3     private $server_name = HOST_DB;
4     private $db_name     = NAME_DB;
5     private $user_name   = USER_DB;
6     private $password    = PASS_DB;
7
8     private $con;
9
10    public function __construct(){
11        // check database connection
12        $this->con = $this->db_connection($this->server_name, $this->user_name, $this->password, $this->db_name);
13        if($this->con==false){
14            // status connection failed.
15        }else{
16            // status connected successfully.
17        }
18    }
19
20    private function db_connection($srvr_nm, $usr_nm, $psswr, $db_nm){
21        // create connection
22        $conn = new mysqli($srvr_nm, $usr_nm, $psswr, $db_nm);
23        // check connection
24        if ($conn->connect_error) {
25            //die("Connection failed: " . $conn->connect_error);
26            return false;
27        }
28        return $conn;
29    }
30
31    public function query($sql){
32        $result = $this->con->query($sql);
33
34        return $result;
35    }
36
37    public function db_close(){
38        $this->con->close();
39    }
40
41 }
```

PHP Hypertext Preprocessor file | length: 938 lines: 42 | Ln: 1 Col: 1 Pos: 1 | Windows (CR LF) | UTF-8 | INS

File #2: /app/model/Lecturer_model.php

```
1 <?php
2 class Lecturer_model{
3     private $db;
4
5     public function __construct(){
6         // create object from database class
7         $this->db = new Database;
8
9         // check status
10        if($this->db == false){
11            echo "<script>console.log('Connection failed.' );</script>";
12        }else{
13            echo "<script>console.log('Connected successfully.' );</script>";
14        }
15    }
16
17    public function getAllDataLecturer(){
18        $result = $this->db->query("select * from tbl_lecturer;");
19        $this->db->db_close(); // Close database connection
20
21        if ($result->num_rows > 0) {
22            // convert to associative array
23            $rows = mysqli_fetch_all($result, MYSQLI_ASSOC);
24            return $rows;
25        } else {
26            return []; // Empty array
27        }
28    }
29
30    public function insertDataLecturer($data){
31        $id = $data['lec_id'];
```

Continue
Code

Continue Code



Lecturer_model.php

```
29
30 public function insertDataLecturer($data) { ②
31     $id = $data['lec_id'];
32     $name = $data['lec_name'];
33     $email = $data['lec_email'];
34     $phone = $data['lec_phone'];
35     $sql = "INSERT INTO tbl_lecturer (id, name, email, phone) VALUES ('$id', '$name', '$email', '$phone')";
36     if($this->db->query($sql) === TRUE) {
37         return true;
38     } else {
39         return false;
40     }
41 }
42
43 public function deleteDataLecturer($id) { ③
44     $sql = "DELETE from tbl_lecturer WHERE id = $id";
45     if($this->db->query($sql) === TRUE) {
46         return true;
47     } else {
48         return false;
49     }
50 }
```

```
51
52 public function updateDataLecturer($data) { ④
53     $id = $data['lec_id_update'];
54     $name = $data['lec_name_update'];
55     $email = $data['lec_email_update'];
56     $phone = $data['lec_phone_update'];
57     $sql = "UPDATE tbl_lecturer SET name='$name', email='$email', phone='$phone' WHERE id='$id'";
58     if($this->db->query($sql) === TRUE) {
59         return true;
60     } else {
61         return false;
62     }
63 }
64 }
65 ?>
```

File #3: /app/controller/Home.php

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

Folder as Workspace

Home.php

```
1 <?php
2 class Home extends Controller{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10        // Associative Arrays (arrays with keys)
11        $arr_data['name'] = $name;
12        $arr_data['age'] = $age;
13        $arr_data['title'] = "Home Page";
14        $arr_data['lecturers'] = $this->logic("Lecturer_model")->getAllDataLecturer();
15        // Display page and send data
16        $this->display('template/header', $arr_data);
17        $this->display("home/index", $arr_data);
18        $this->display('template/footer');
19    }
20
21    public function page($current_page=2, $next_page=3, $prev_page=1){
22        // Associative Arrays (arrays with keys)
23        $arr_data['current'] = $current_page;
24        $arr_data['next'] = $next_page;
25        $arr_data['previous'] = $prev_page;
26        $arr_data['title'] = "Personal Page";
27        // Display page and send data
28        $this->display('template/header', $arr_data);
29        $this->display("home/page", $arr_data);
30        $this->display('template/footer');
31    }
32 }
```

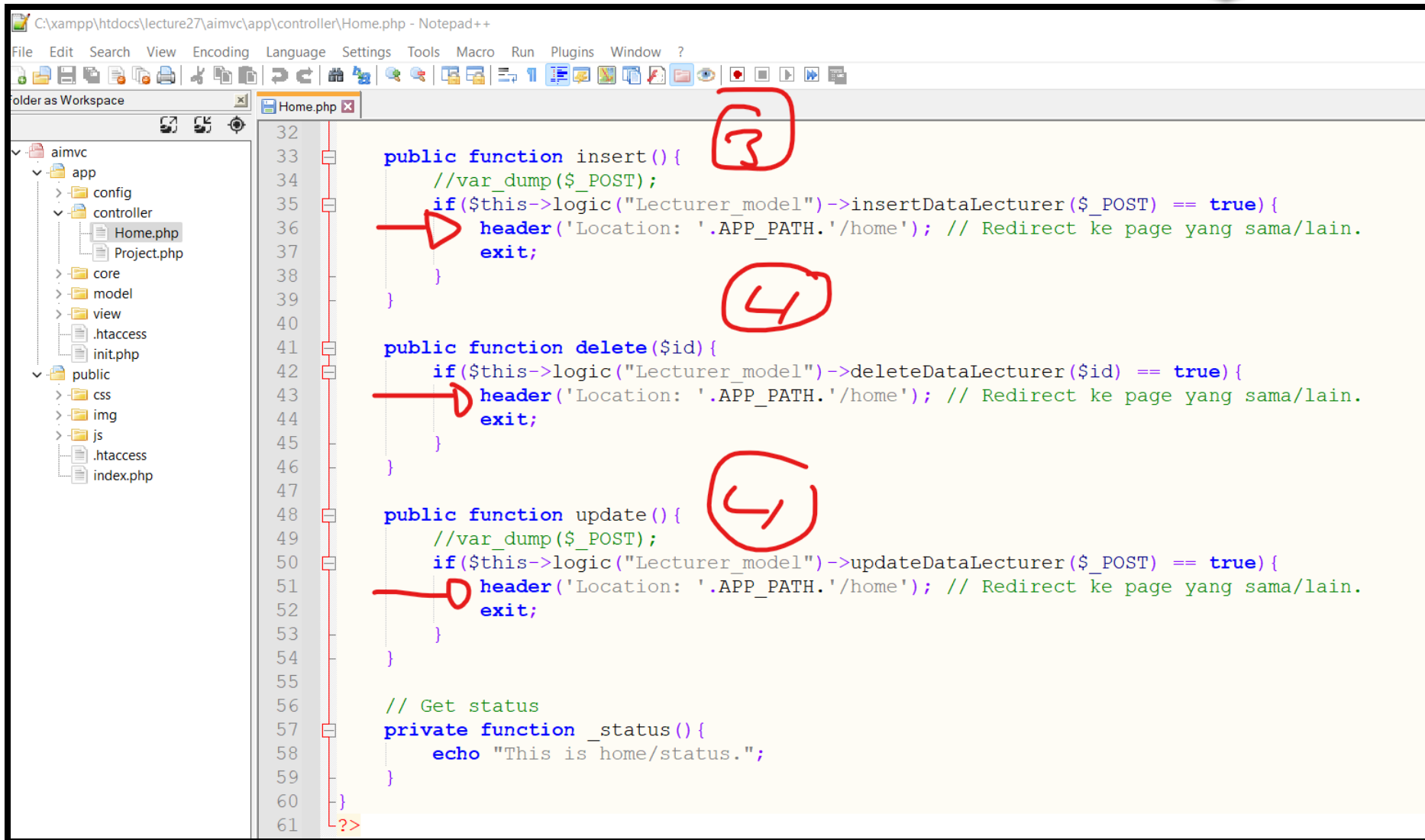
1

2

Continue Code

After insert, update or delete data, the URL will be redirected to the same page.

**Continue
Code**



```
C:\xampp\htdocs\lecture27\aimvc\app\controller\Home.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
older as Workspace
Home.php
32
33 public function insert(){
34     //var_dump($_POST);
35     if($this->logic("Lecturer_model")->insertDataLecturer($_POST) == true){
36         header('Location: '.APP_PATH.'/home'); // Redirect ke page yang sama/lain.
37         exit;
38     }
39 }
40
41 public function delete($id){
42     if($this->logic("Lecturer_model")->deleteDataLecturer($id) == true){
43         header('Location: '.APP_PATH.'/home'); // Redirect ke page yang sama/lain.
44         exit;
45     }
46 }
47
48 public function update(){
49     //var_dump($_POST);
50     if($this->logic("Lecturer_model")->updateDataLecturer($_POST) == true){
51         header('Location: '.APP_PATH.'/home'); // Redirect ke page yang sama/lain.
52         exit;
53     }
54 }
55
56 // Get status
57 private function _status(){
58     echo "This is home/status.";
59 }
60 }
61 ?>
```

The screenshot shows a Notepad++ editor with the file `C:\xampp\htdocs\lecture27\aimvc\app\controller\Home.php`. The code defines three public functions: `insert()`, `delete($id)`, and `update()`. Each function calls a logic method to perform a database operation. If the operation is successful (returns `true`), the code uses `header('Location: '.APP_PATH.'/home');` to redirect the user to the home page, followed by `exit;`. Red annotations highlight the `header` and `exit` statements in each function. A red arrow points to the `header` statement in the `insert` function, and another red arrow points to the `header` statement in the `delete` function. The `update` function also has a red arrow pointing to its `header` statement. The code is enclosed in a red box, and the `header` and `exit` statements are circled in red. The `insert` function is labeled with a red '3' and the `delete` function is labeled with a red '4'.

File #4: /app/view/home/index.php

Design user interface, table and buttons

Modifikasi button “Edit” di bawah ini dengan menggantikan code pada button tag dengan code pada kotak warna merah.

```
1 <div class="container">
2   <h1>This is my Home Page.</h1>
3
4   <p>I'm <?php echo $data['name']; ?> from Manado.</p>
5   <p>I'm <?php echo $data['age']; ?> years old this year.</p>
6
7   <h3>Daftar Dosen:</h3>
8   <button type="button" class="btn btn-success" data-bs-toggle="modal" data-bs-target="#formInsertBox">Add New Lecturer</button>
9   <table class="table table-hover">
10    <thead>
11     <tr>
12      <th class="text-center">#</th>
13      <th class="text-center">ID</th>
14      <th class="text-center">Name</th>
15      <th class="text-center">E-mail</th>
16      <th class="text-center">Phone Number</th>
17      <th class="text-center">Actions</th>
18     </tr>
19    </thead>
20    <tbody>
21     <?php
22      $count = 1;
23      foreach($data['lecturers'] as $lecs): ?>
24       <tr>
25        <td class="text-center"><?php echo $count; ?></td>
26        <td class="text-center"><?php echo $lecs['id']; ?></td>
27        <td><?php echo $lecs['name']; ?></td>
28        <td><?php echo $lecs['email']; ?></td>
29        <td class="text-center"><?php echo $lecs['phone']; ?></td>
30        <td class="text-center">
31         <button class="btn btn-info" data-id="<?php echo $lecs['id']; ?>" onclick="$('#lec_id_update').val('<?php echo $lecs['id']; ?>'); $('#lec_name_update').val('<?php echo $lecs['name']; ?>'); $('#lec_email_update').val('<?php echo $lecs['email']; ?>'); $('#lec_phone_update').val('<?php echo $lecs['phone']; ?>'); $('#formUpdateBox').modal('show');" >Edit</button>
32         <a href="<?php echo APP_PATH; ?>/home/delete/<?php echo $lecs['id']; ?>" type="button" class="btn btn-danger">Delete</a>
33        </td>
34       </tr> <?php $count++; ?>
35     </tbody>
36    </table>
37    
38  </div>
39
40  <!-- Insert Modal -->
41  <div class="modal" id="formInsertBox">
42    <div class="modal-dialog modal-dialog-centered">
43      <div class="modal-content">
44        <!-- Modal Header -->
45        <div class="modal-header">
46          <h4 class="modal-title">Insert New Data Lecturer</h4>
47          <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
48        </div>
```

Handwritten notes and annotations:

- A red box highlights the `<button>` tag in the code.
- A green box contains the replacement code for the `<button>` tag.
- A red arrow points from the green box to the `<button>` tag in the code.
- Handwritten text "Column" with a bracket indicates the table structure.
- A circled number "2" is next to the `<td>` tag containing the `<button>` tag.

Continue
Code

Bootstrap's Centered Modal Box for Input Data

```
index.php x
40
41 <!-- Insert Modal -->
42 <div class="modal" id="formInsertBox">
43   <div class="modal-dialog modal-dialog-centered">
44     <div class="modal-content">
45       <!-- Modal Header -->
46       <div class="modal-header">
47         <h4 class="modal-title">Insert New Data Lecturer</h4>
48         <button type="button" class="btn-close" data-bs-dismiss="modal"></button>
49       </div>
50       <!-- Modal body -->
51       <div class="modal-body">
52         <form action="<? = APP_PATH;?>/home/insert" method="post">
53           <div class="mb-3 mt-3"> <label for="lec_id">ID:</label>
54             <input type="number" class="form-control" id="lec_id" placeholder="Enter ID" name="lec_id">
55           </div>
56           <div class="mb-3 mt-3"> <label for="lec_name">Lecturer Name:</label>
57             <input type="text" class="form-control" id="lec_name" placeholder="Enter Name" name="lec_name">
58           </div>
59           <div class="mb-3 mt-3"> <label for="lec_email">E-mail:</label>
60             <input type="email" class="form-control" id="lec_email" placeholder="Enter E-mail" name="lec_email">
61           </div>
62           <div class="mb-3"> <label for="lec_phone">Phone Number:</label>
63             <input type="number" class="form-control" id="lec_phone" placeholder="Enter Phone Number" name="lec_phone">
64           </div>
65         </div>
66       <!-- Modal footer -->
67       <div class="modal-footer">
68         <button type="button" class="btn btn-danger" data-bs-dismiss="modal">Cancel</button>
69         <button type="submit" class="btn btn-danger">Insert Data</button>
70       </div>
71     </form>
72   </div>
73 </div>
74 </div>
```

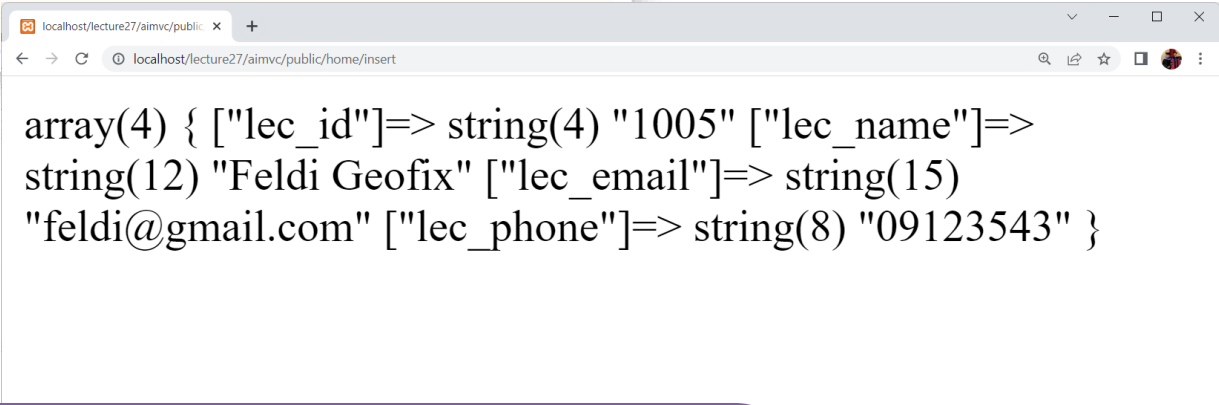
Continue
Code



Parameter Sent *after* Click Button Insert

Cari fungsi *insert()* di bawah ini, dan pastikan *var_dump* tidak di comment supaya bisa melakukan *testing* untuk melihat apakah data yang tepat sudah di kirim ke server.

```
public function insert(){  
    var_dump($_POST);  
    //if($this->logic("Lecturer_model")->insertDataLecturer($_POST) == true){  
    //    header('Location: '.APP_PATH.'/home'); // Redirect ke page yang sama/lain.  
    //    exit;  
    //}  
}
```



```
array(4) { ["lec_id"]=> string(4) "1005" ["lec_name"]=>  
string(12) "Feldi Geofix" ["lec_email"]=> string(15)  
"feldi@gmail.com" ["lec_phone"]=> string(8) "09123543" }
```

Ini adalah parameter yang dikirim dengan menggunakan method POST.

```
["lec_id"]=> string(4) "1005"  
["lec_name"]=> string(12) "Feldi Geofix"  
["lec_email"]=> string(15) "feldi@gmail.com"  
["lec_phone"]=> string(8) "09123543"
```



Postman is an app that allows for web API testing. Why use postman for API testing?

Postman adalah platform kolaborasi untuk pengembangan API. Fitur Postman menyederhanakan setiap langkah pembuatan API dan menyederhanakan kolaborasi sehingga kita dapat membuat API yang lebih baik dan lebih cepat.

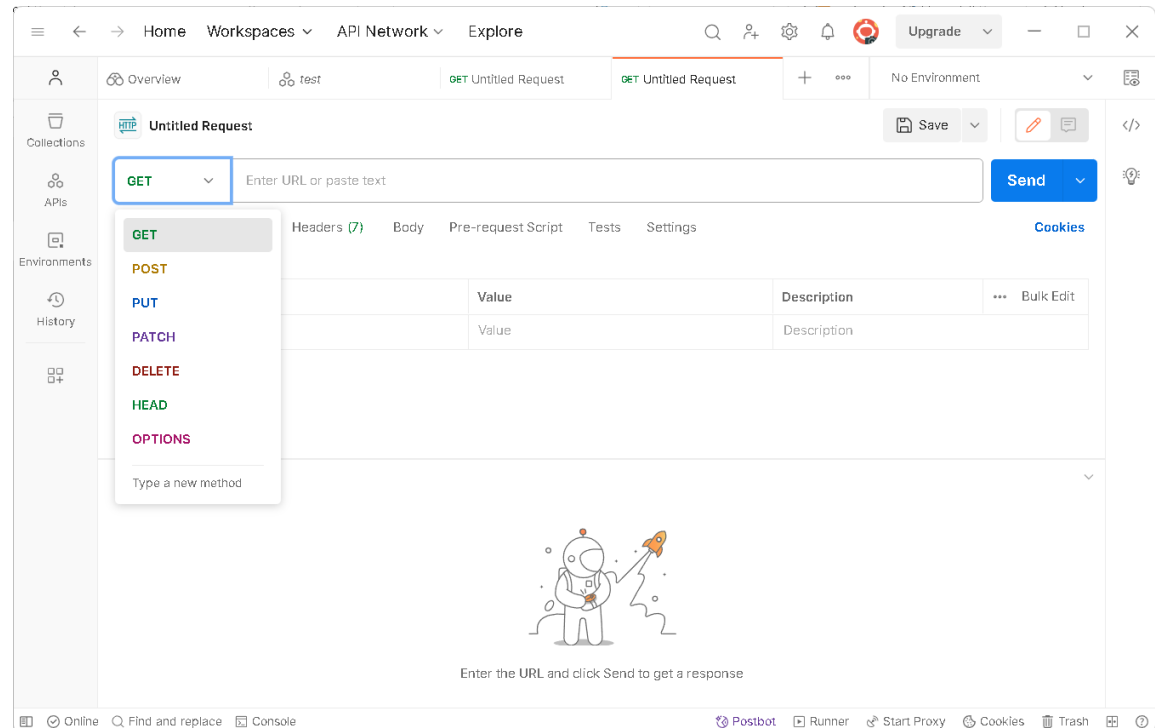
Gunakan URL API yang sudah dibuat (Contoh):

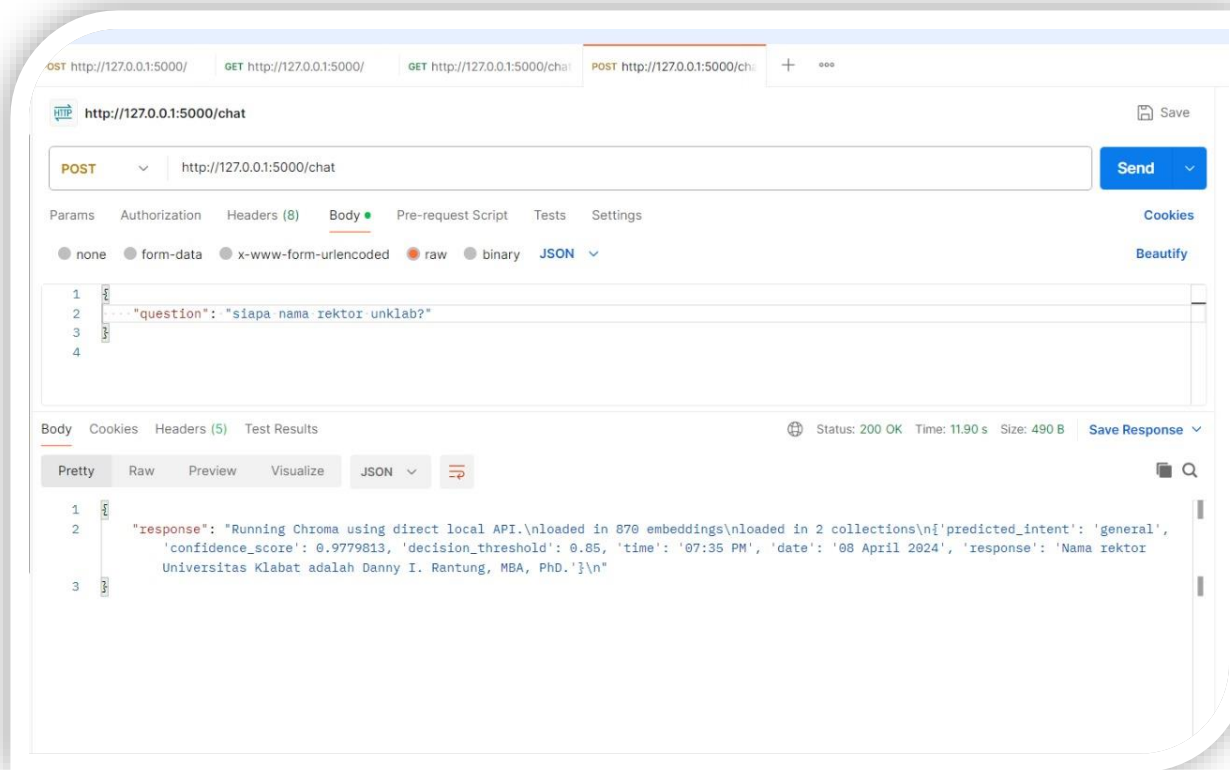
<http://localhost/lecture27/aimvc/public/home/insert/>

Cara download Postman:

- 1) Buka official website postman di <https://www.postman.com/downloads/>
- 2) Pilih Download App
- 3) Selesai Download Double klik dan pilih install -> Tunggu sampai proses selesai.
- 4) Jika sudah selesai silahkan klik tombol Launch untuk menjalankan
- 5) Silahkan Registrasi menggunakan email aktif

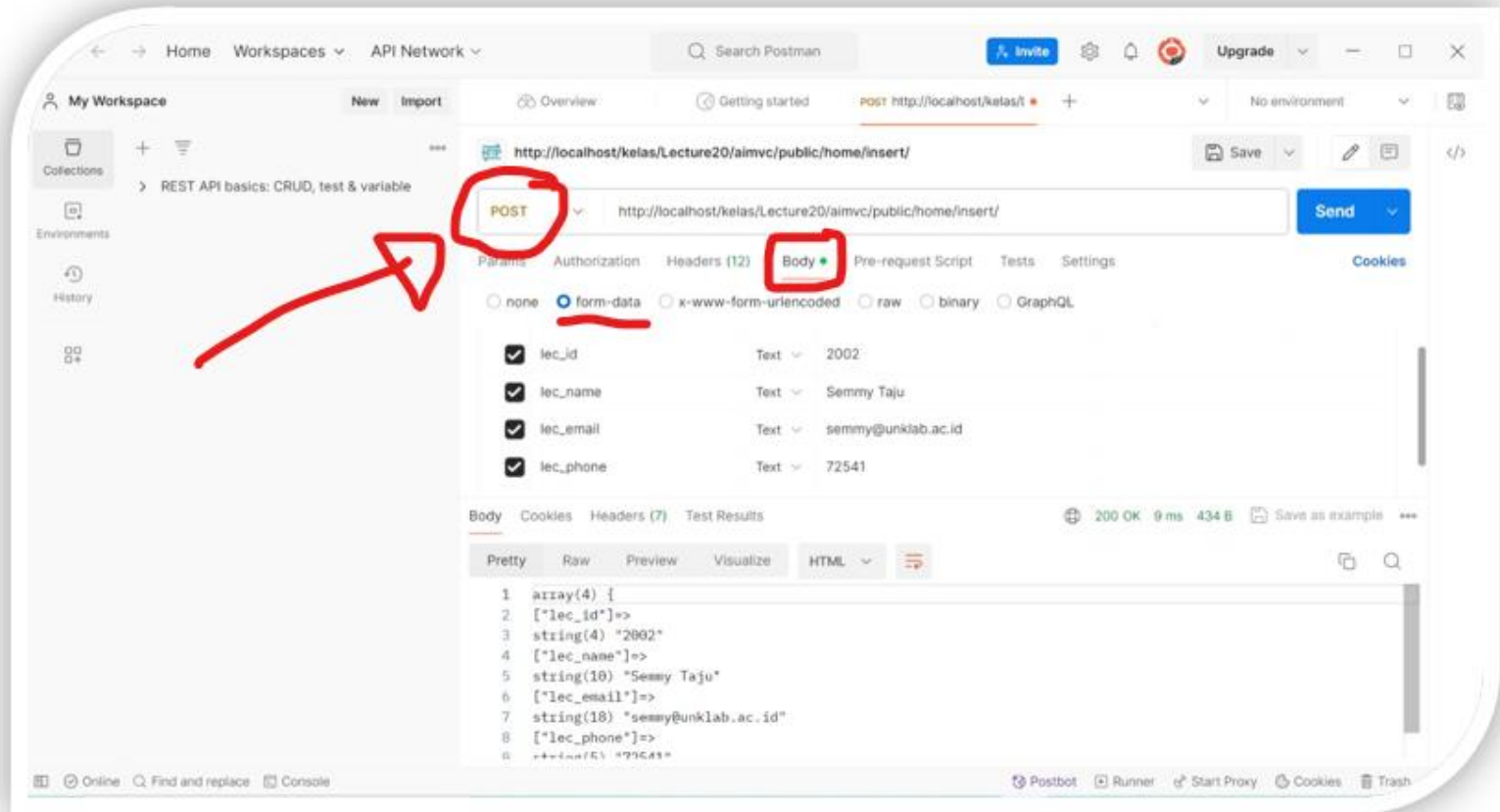
**Sign in dan
Silahkan
Demo cara
menggunakan
Postman:**





API testing is crucial for ensuring reliability and preserving consumer trust. **Postman** is a popular tool for API testing that allows developers to create and execute HTTP requests and test API responses.

```
public function insert(){  
    var_dump($_POST);  
    //if($this->logic("Lecturer_model")->insertDataLecturer($_POST) == true){  
    //    header('Location: '.APP_PATH.'/home'); // Redirect ke page yang sama/lain.  
    //    exit;  
    //}  
}
```



Jangan lupa untuk memasukkan input data yang akan di kirimkan ke sisi server dengan menggunakan methode POST.

A decorative border of green leaves and branches at the top of the slide.

6. Demo Functions

(Insert, Update & Delete Data)

A decorative border of green leaves and branches at the bottom of the slide.

View All Data *in* Table

Home Page

localhost/lecture27/aimvc/public/home

AI Home Page Project

This is my Home Page.

I'm Juan from Manado.

I'm 24 years old this year.

Daftar Dosen:

Add New Lecturer

#	ID	Name	E-mail	Phone Number	Actions	
1	1001	Jhon Tegar	john@example.com	098121213	Edit	Delete
2	1002	Deti Tibar	deti@example.com	098121234	Edit	Delete
3	1003	Dony Desas	dony@example.com	098121345	Edit	Delete
4	1004	Ferel Sanger	ferel@yahoo.com	0912345678	Edit	Delete



Insert Function

Home Page

localhost/lecture27/aimvc/public/home

AI Home Page Project

This is my Home Page

I'm Juan from Manado.

I'm 24 years old this year.

Daftar Dosen:

Add New Lecturer

#	ID	Name
1	1001	Jhon Tegar
2	1002	Deti Tibar
3	1003	Dony Desas
4	1004	Ferel Sanger

Actions

Edit Delete

Edit Delete

Edit Delete

Edit Delete

Insert New Data Lecturer

ID:

Lecturer Name:

E-mail:

Phone Number:

Cancel Insert Data



Update Function

The screenshot shows a web browser window with the address `localhost/lecture27/aimvc/public/home`. The page has a navigation bar with 'AI', 'Home', 'Page', and 'Project'. The main content area includes a header 'This is my Home', a bio 'I'm Juan from Manado. I'm 24 years old this year.', and a section titled 'Daftar Dosen:' with a green 'Add New Lecturer' button. Below this is a table of lecturers. A modal window titled 'Update Data Lecturer' is open, displaying a form with fields for ID (1002), Lecturer Name (placeholder: Enter Name), E-mail (placeholder: Enter E-mail), and Phone Number (placeholder: Enter Phone Number). At the bottom of the modal are 'Cancel' and 'Update Data' buttons. In the background, the 'Actions' column of the lecturers table shows 'Edit' and 'Delete' buttons for each row.

#	ID	Name
1	1001	Jhon Tegar
2	1002	Deti Tibar
3	1003	Dony Desas
4	1004	Ferel Sanger

Actions	
Edit	Delete
Edit	Delete
Edit	Delete
Edit	Delete



Solve it

Open Innovation
Challenge

+ Add User				
Book Title	Author	Category	Price	Actions
Romeo And Juliet	William Shakespeare	Tragedy	100	Edit Delete
The Lord of the Rings	J. R. R. Tolkien	Classic	69.99	Edit Delete
The Two Towers	J. R. R. Tolkien	Classic	69.99	Edit Delete
The Fellowship of the Ring	J. R. R. Tolkien	Classic	69.99	Edit Delete
Little Women	May Alcott	Classic	50.99	Edit Delete
The Chronicles of Narnia	C. S. Lewis	Fiction	39.99	Edit Delete

Challenge #1

Tambahkan/Buatkan *style UI* baru versi kalian sendiri sehingga tampilan dari sistem CRUD yang dibuat semakin bagus untuk di lihat.



Challenge #2

Tambahkan *real data/information* di dalam database anda dengan memasukan paling kurang **15 data dosen** di UNKLAB.

The screenshot shows a web application interface for 'Universitas Klabat'. A modal window titled 'Attendance Schedule' is open, displaying a form for creating a new attendance schedule. The form includes the following fields:

- Course Topic:** A text input field.
- Course:** A text input field.
- Meeting:** A text input field.
- Lecturer Email:** A text input field with a 'Clear' button.
- Date:** A date picker showing 'dd/mm/2024'.
- Time:** A text input field with the placeholder 'Waktu presensi' and a 'Select Options' dropdown.
- Max Radius:** A text input field with the placeholder 'Input jarak' and a 'Select Options' dropdown.

At the bottom of the modal, there are two buttons: 'Cancel' and 'Create Schedule'. A calendar for April 2024 is open, showing the 17th as the selected date.

Challenge #3

Buatlah sebuah kolom baru (“*date*”) di table *tbl_lecturer* dan tambahkan input date pada fungsi insert data dosen yang telah dibuat.

Login Form

Login Signup

Email Address

Password

[Forgot password?](#)

Login

Not a member? [Signup now](#)

Challenge #4

Buatlah sebuah controller dan view yang baru untuk menangani Login dan Signup (form) dari lebih dari 1 users. Buat juga table baru *tbl_login* dan beberapa baris data untuk demo login multiple users nantinya.

[PERTAMA]

**Lakukan uji coba di
local PC masing-
masing.**

Latihan untuk *Students*

Create directory : C:\xampp\htdocs\lecture27\
Create file : `_.php`

Buat semua contoh yang sudah di pelajari sebelumnya di *lecture* ini.

[SELANJUTNYA]

**Upload code Lecture #27
ke sisi server untuk di
DEMO.**

Create Directory *in* Server

[1] Masuk sebagai *administrator* atau *root*:

➤ ***sudo su***

[2] Buat project folder baru dengan perintah (Parent directory):

➤ ***sudo mkdir -p /var/www/html/lecture27***

[3] Konfigurasi permisi hak akses dan pemilik dari folder:

➤ ***sudo chmod -R 755 /var/www/html/lecture27***

➤ ***sudo chown -R semmy:semmy /var/www/html/lecture27***

[4] Restart Apache Web Server:

➤ ***sudo systemctl restart apache2***

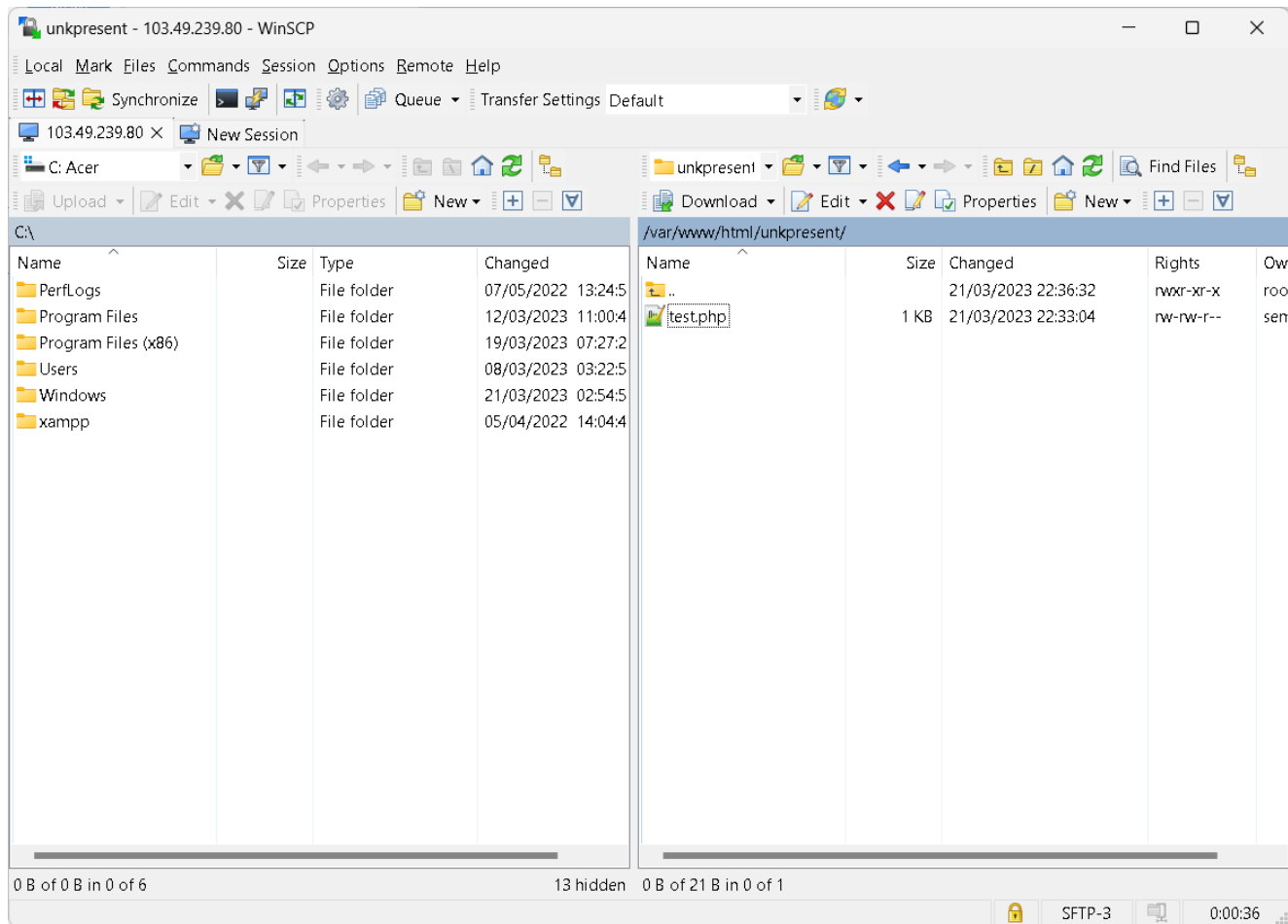
Note:

Ubah nama user “**semmy**” dengan nama *user* access ubuntu kalian.

```
root@unklab: /var/www/html
root@unklab:/home/semmy# sudo mkdir -p /var/www/html/unkpresent
root@unklab:/home/semmy# sudo chmod -R 755 /var/www/html/unkpresent
root@unklab:/home/semmy# cd ..
root@unklab:/home# cd ..
root@unklab:/# ls
bin  dev  home  lib32  libx32  media  opt  root  sbin  srv  tmp  var
boot  etc  lib  lib64  lost+found  mnt  proc  run  snap  sys  usr
root@unklab:/# cd var
root@unklab:/var# ls
backups  crash  local  log  opt  snap  tmp
cache  lib  lock  mail  run  spool  www
root@unklab:/var# cd www/
root@unklab:/var/www# ls
html
root@unklab:/var/www# cd html/
root@unklab:/var/www/html# ls
index.html  test.php  unkpresent
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 root root 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html# sudo chown -R semmy:semmy /var/www/html/unkpresent
root@unklab:/var/www/html# sudo systemctl restart apache2
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 semmy semmy 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html#
```

Upload Project via WinSCP

- 1) Buka WinSCP dan arahkan ke “/var/www/html/lecture27”.
- 2) Selanjutnya, upload **file code lecture#27** dari local PC dan ke directory server menggunakan WinSCP.
- 3) Buka via Browser.



Output ditampilkan menggunakan domain name, HTTPS dan bisa di akses secara public via Browser:

Contohnya:

<https://domainkelompok.site/lecture27/aimvc/public/>

Kemungkinan Error
pada waktu di upload
ke server Ubuntu.

Create Password *for* Root

Pastikan **root user** harus memiliki password di MySQL database. Dengan mengetikan perintah berikut:

➤ ***sudo mysql***

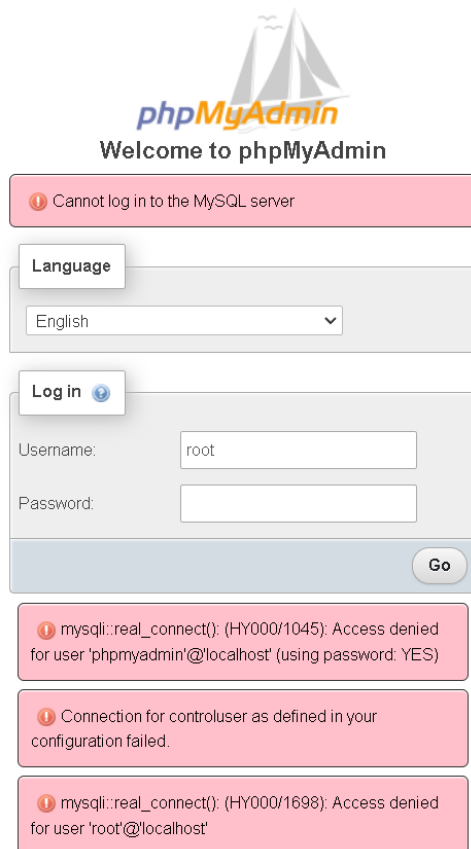
➤ **mysql>** ALTER USER 'root'@'localhost' IDENTIFIED WITH mysql_native_password BY '**ganti_password_kita**';

Kemudian keluar dari MySQL prompt
mysql> exit

Create New User DB

Buat user yang baru, jangan gunakan root.

Error: mysqli::real_connect(): (HY000/1045): Access denied for user 'phpmyadmin'@'localhost' (using password: YES)



The screenshot shows the phpMyAdmin login interface. At the top, there's a logo and the text "Welcome to phpMyAdmin". Below this, a pink error box states: "Cannot log in to the MySQL server". There's a "Language" dropdown menu set to "English". A "Log in" button is next to the login fields. The "Username:" field contains "root" and the "Password:" field is empty. A "Go" button is at the bottom right of the login form. Below the login form, there are three more pink error boxes: 1. "mysqli::real_connect(): (HY000/1045): Access denied for user 'phpmyadmin'@'localhost' (using password: YES)" 2. "Connection for controluser as defined in your configuration failed." 3. "mysqli::real_connect(): (HY000/1698): Access denied for user 'root'@'localhost'"

Change to super user:

➤ `sudo su`

Access MySQL:

➤ `mysql -u root -p`

Create User phpmyadmin with Password:

- `CREATE USER 'testadmin'@'localhost' IDENTIFIED BY '1234567890';`
- `GRANT ALL ON testadmin.* TO 'testadmin'@'localhost';`
- `GRANT ALL PRIVILEGES ON *.* TO 'testadmin'@'localhost' WITH GRANT OPTION;`
- `FLUSH PRIVILEGES;`
- `Exit`
- Access Again MySQL:**
 - `mysql -u testadmin -p`
 - Selanjutnya, buat database
 - Buat table

How to solve 'Fatal error': Class 'MySQLi' not found?

Install these packages:

- `sudo apt-get install php-mysqlnd`
- `sudo apt-get install php5-mysql`
- `sudo apt-get install php-mysql`

Restart Apache2:

- `sudo service apache2 restart`

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

